Project Management Protocol of the Netherlands eScience Center

Authors: Netherlands eScience Center

Date: July 2026
Version: 4.0

Date	Version	Changes
1 July 2026	4.0	Major changes, renew protocol after 2025 restructuring.
1 June 2025	3.5	redacted internal references, added RSD checklist appendix and missing intro text, included Zenodo references, fixed tables, and improved description of annual review meeting
1 December 2024	3.0	Planning, Legal agreements, Project budget breakdown, change in the workshops and end report processes, open-access publications
1 October 2023	2.0	DT and PM mandates, Roles of Directors, PM role in Ambition 2, SS, KD and external projects, Communications role, Editorial team, breakdown of hours, opportunities beyond project, Technology Status Report, End report
23 September 2022	1.0	initial version

Contents

1 Scope and definitions **3**

1.1 Scope . . . 3

1.2 Types of projects . . . 3

1.3 Stakeholders . . . 4

1.4 Project lifecycle overview . . . 7

2 Project Initiation **8**

2.1 SH assignment . . . 8

2.2 Project registration . . . 8

2.3 Project staffing . . . 9

2.4 Administrative kick-off meeting . . . 9

2.5 Kick-off meeting . . . 10

2.6 Legal agreements . . . 11

2.7 Technology plan . . . 12

3 Project execution **14**

3.1 Project logging . . . 14

3.2 Status update meetings . . . 14

3.3 Project team meetings . . . 15

3.4 Writing hours and managing project budget . . . 15

3.5 Workshops . . . 16

3.6 Data and Software Management Plans . . . 17

3.7 Knowledge transfer . . . 17

3.8 Code quality and sustainability checks . . . 19

3.9 Annual project review meeting . . . 20

3.10 Reporting . . . 22

3.11 Conflict resolution and complaint procedure . . . 23

3.12 Non-RSE Consultants . . . 23

3.13 Changes to the project . . . 23

3.14 Opportunities beyond the project . . . 25

4 Project closing **26**

4.1 Handling end report . . . 26

4.2 Formally closing the project . . . 27

Appendices **30**

A Applying the protocol to external projects **30**

B Applying the protocol to internal projects 31

C Lead RSE role description 32

D Example of the project log 33

E RSD project pages 34

F Internal Procedure of the Project Workshops Approval 35

Key eScience Roles appearance 36

1 Scope and definitions

1.1 Scope

This document is the official project management protocol of the Netherlands eScience Center. It describes the activities, responsibilities, and procedures required to manage an eScience project after it has been formally awarded, through to its formal closure.

The protocol primarily applies to projects awarded through the eScience Center calls for proposals, although it also provides guidance for other types of projects as defined in Section 1.2. It follows the project life cycle and, for each phase, specifies the required activities, optional activities, responsible roles, and expected outcomes.

The call process itself is governed by a separate protocol [1] and is therefore outside the scope of this document. This protocol starts once a project has been formally awarded and covers all activities until the project is formally closed.

The following topics are also outside the scope of this protocol:

- independent project evaluation (impact, outputs, process, and collaboration);
- detailed guidance on project management tools (e.g. Ganttic, Metabase dashboards, AFAS);
- general project management methodologies and techniques.

The Management Team (MT) approves this protocol and reviews it annually. The document is organised according to the project life cycle (Section 1.4), presenting activities in chronological order.

1.2 Types of projects

The eScience Center receives an annual budget from NWO and SURF , the larger part of which is allocated to projects submitted by researchers working at eligible research performing organizations in the Netherlands in the form of the in-kind provision of Research Software Engineers (RSEs). The eScience Center also participates in externally funded projects and carries out internally funded strategic initiatives.

These projects differ in their funding model and governance. Throughout this protocol, they are grouped into the categories shown in Table 1.

Table 1: Project categories covered by this protocol

Project type	Description	Governed by
Call projects	Projects funded through the Netherlands eScience Center calls for proposals and delivered through in-kind RSE support.	This protocol
External projects	Projects funded by external organisations (e.g., NWO, Horizon Europe, industry). External contractual obligations take precedence where applicable.	This protocol + external agreements. See Appendix A for further details.
Other projects	Internal initiatives such as Research and Development (R&D) projects, Fellowship projects, and Dissemination and Community (D&C) initiatives.	This protocol where applicable + dedicated internal procedures. See Appendix B for further details.

1.3 Stakeholders

An eScience project is a project involving the eScience Center, where responsibility is shared between different stakeholders who each have their own roles and responsibilities during specific phases of the project.

Stakeholder	Abbreviation	Assignment	Role	More info
Lead Applicant	LA	main applicant and recipient of the grant	primary contact for the eScience Center project, accountable for the (quality of the) scientific contribution to the project	responsibilities defined in the call text, Terms and Conditions, and possibly a Consortium/Collaboration agreement.
Programme Manager	PM	assigned by the PM team [1]	accountable for the eScience contribution, delivering results within scope, time, and budget; project budget holder	Full text of responsibilities available in the PM job profile document and PM mandate (see [2])
Lead Research Software Engineer	**Lead RSE**	appointed by accountable PM	responsible for the timely execution of the project, main contact person for the project with other stakeholders	More details on responsibilities, see the formal role description of **Lead RSE** (Appendix C).
Research Software Engineer (assigned to a project)	RSE	assigned by the accountable PM	responsible for the timely completion of the project	Job, role descriptions given in [3, 4]. All RSE activities coordinated by **Lead RSE** in agreement with PM.
Consulting Research Software Engineer	Consulting RSE	involved at request of **Lead RSE** or accountable PM	provides expertise to the project for a defined period; may also join key meetings in addition to an expertise contribution	All activities coordinated by **Lead RSE** in agreement with PM in case the project team needs additional expertise.
Technology Lead.	TL	assigned by the TL team [2]	acts as point of contact for **Lead RSE** to the TL team. Safeguards the technological aspects of a project; accountable for the quality, reuse and sustainability of the research software developed.	The TLs team is responsible for internal training programme of RSEs.
Section Head	**SH**	assigned by the PM/SH team [3]	line manager of RSEs, responsible for monitoring the overall effectiveness of RSEs in bringing projects to completion; maintain overview of a research domain.	The **SH** team assigns one **SH** to each RSE team, and the **SH** ensures that team keeps its capacity and planning up to date.
Communications			advise and support internal and external project communication, including showcasing through news, website, newsletters, social media, interviews, and videos.	

Stakeholder	Abbreviation	Assignment	Role	More info
Community Manager	CM		advise on developing outreach activities and promoting community engagement, responsible for external training programme	
Secretary			organizes formal meetings, provide with agenda and slide template, invitation text, list of participants (with emails), and timeline.	
Programme Director	PD		the escalation point for PMs and liaison to DT for project-related decisions; approves workshop plans (with Finance) and holds the Acquisition budget.	
Director of Technology	DoT		the escalation point for TLs, the contact point of TLs to DT, accountable (and responsible) for licences and Intellectual Property (IP), software sustainability budgets holder	
Director of Operations	DoO		handles legal questions (e.g., contracts, Collaborative Agreements and guest agreements)	
Finance & Control	Finance	part of Operations, includes Controller, and led by DoO	responsible for maintaining finances and project administration	
Directors Team	DT	comprised of DoT, DoO, General Director and PD	approves formal decisions regarding projects (e.g., budget changes)	
GDPR contact person		appointed by the DT, see the Intranet for contact information	consults on GDPR/AVG [5, 6, 7] or privacy-related issues in the project	The eScience Center has not appointed a Data Protection Officer. GDPR aspects must be discussed with the contact person.
eScience Center project team	eScience project team	comprises RSEs, PM and TL working on the project	responsible for the timely completion of the project	

Stakeholder	Abbreviation	Assignment	Role	More info
Project team		comprises the eScience project team, LA and their team (including team members indicated in the project proposal)	responsible for the timely completion of the project	
Editorial Team			provides support with outreach	The eScience Center maintains a blog, and has presence in major social networks

[1] All PMs, led by the Programme Director, constitute the PM team.

[2] All TLs, led by Director of Technology, constitute the TL team.

[3] All **SH**, led by Executive Director, constitute the **SH** team.

1.4 Project lifecycle overview

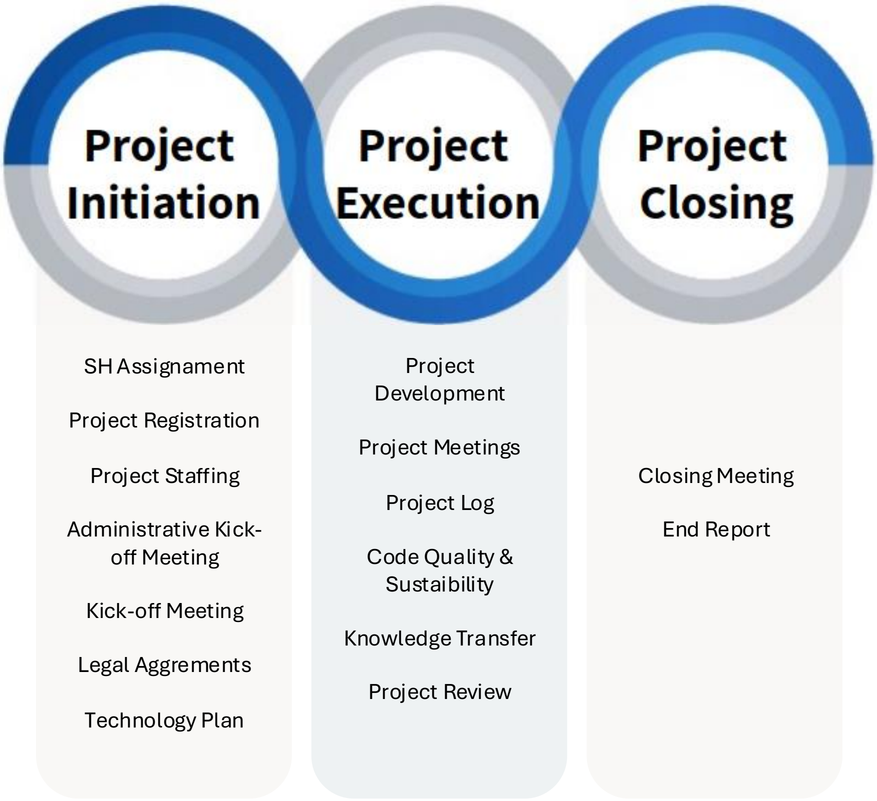

Figure 1: Project Lifecycle Overview

At the eScience Center, a standard project lifecycle consists of three phases (see Figure 1) initiated, followed by the execution phase, during which research software is developed and project activities are coordinated and monitored. Finally, the project is formally closed through a structured closing process.

The following sections describe each phase of the project lifecycle in detail.

2 Project Initiation

The project initiation phase begins once a project has been approved for funding. Following the communication of the awarded project by the Program Officer (PO), the responsible **SH** initiates the project setup. The purpose of this phase is to establish the project within the eScience Center by completing the required administrative setup, assigning responsibilities, and preparing the project for execution.

The exact initiation process depends on the project type. External projects, as well as certain internal projects, may require additional contractual or legal arrangements before they can commence. Further information on the different project types is provided in Appendix B and Appendix A.

Once the project has been registered, staffed, and all applicable administrative, financial, and legal requirements have been fulfilled, it is ready to enter the execution phase.

2.1 SH assignment

Each project has one assigned **SH**, who is accountable for the project throughout its lifecycle.

For projects funded through the eScience Center's regular funding calls, the assigned **SH** normally corresponds to the section under which the proposal was submitted. In other cases, or where the assignment is not self-evident, the Section Heads jointly determine the most appropriate assignment, taking into account the project's scope, expertise, and organizational responsibilities.

Once the assignment has been agreed, the responsible **SH** is recorded and Finance is notified to initiate the project registration process.

Responsible: Section Heads.

2.2 Project registration

The table below summarizes the main project registration outputs, the responsible stakeholder, and the associated activities.

Table 3: Project registration activities and responsibilities

Output	Responsible	Main actions
Project record	Finance	• Creating the project code in AFAS • Assigning the responsible **SH** as budget holder
Project portfolio	Finance	• Creating the project portfolio folder [8] following the rquired template sub-folders • Requesting missing project documentation • Uploading the grant package (proposal, awarding letter, signed start form, and supporting documentation submitted by the LA)

Responsible: Finance.

2.3 Project staffing

The responsible **SH** is responsible for staffing the project, including the appointment of a **Lead RSE** and the assembly of the eScience Center project team [9]. Staffing decisions take into account the project's requirements, available capacity, and the expertise required for successful delivery. Staffing may be adjusted throughout the project as needed (see Section 3.13). The **Lead RSE** serves as the primary point of contact for the LA, unless otherwise agreed.

2.3.1 Staffing preparation

The responsible **SH** reviews the project proposal to determine the staffing requirements for the project. The assessment considers the project scope and objectives, the awarded budget, the expected timeline, the required scientific and technical expertise (making use of available organizational information on technology capabilities, e.g., technology skills surveys), any applicable contractual, legal, or organizational constraints, and the available capacity within the section based on the organizational planning system. These considerations form the basis for assigning the Science Center project team.

2.3.2 Team assignament

Every project must have: one accountable **Lead RSE**; and at least one additional Research Software Engineer assigned to the project. The eScience Center project team is assigned according to the following process:

1. The responsible **SH** identifies a suitable candidate for the **Lead RSE** role, taking into account the staffing preparation described above.
2. The **SH** discusses the proposed assignment with the candidate.
3. The proposed staffing is reviewed and confirmed by the Section Heads.
4. The responsible **SH** formally assigns the **Lead RSE** and the remaining project team members.

2.3.3 Project planning

Following project registration, the project is synchronized to the organization's planning system. The responsible **SH** records the **Lead RSE** in the corresponding project in Ganttic. The **Lead RSE** prepares the initial project planning by allocating the assigned team members across the different project phases. The planning serves as the baseline for project execution and may be updated during the project as required.

Responsible: SH.

2.4 Administrative kick-off meeting

Once the project has been registered and staffed, the responsible **SH** organizes an administrative kick-off meeting with the LA. Prior to the meeting, the **SH** and **Lead RSE** review the project proposal and identify any project-specific requirements that should be discussed with the LA. The purpose of the meeting is to introduce the collaboration with the Netherlands eScience Center and to ensure that the project is administratively and organizationally ready to start. These may include:

- any additional organizational or technical support required for the project;
- deviations from the default intellectual property or software licensing arrangements;
- data readiness, data access requirements, and any personal data or GDPR requirements;

- software sustainability expectations; and
- ethical, legal, or other project-specific risks.

The meeting is supported by a standardized administrative kick-off presentation available in the project portfolio under the *Templates projects* folder [8]. The presentation is periodically updated to reflect the requirements of different funding calls and the eScience Center's current procedures and policies. The responsible **SH** uses this presentation to ensure that all standard topics are covered consistently across projects while adapting the discussion to the specific needs of the project.

The presentation typically covers the following topics:

- Introduction to the Netherlands eScience Center (mission, organization, and expertise);
- Working with the eScience Center (roles and responsibilities, collaboration model, in-kind and in-cash contributions);
- Project lifecycle and project governance;
- Community activities and research impact (Research Software Directory, Digital Skills Programme, Fellowship Programme, etc.);
- Intellectual property, software licensing, publication practices, and open science principles;
- Project-specific needs, including scientific and technical expertise, SURF infrastructure requirements, and additional resources (e.g., AI subscriptions or other project-specific services).

The presentation used during the meeting and any meeting notes are stored in the corresponding project portfolio folder together with the other project documentation. The Secretary may assist the **SH** with scheduling the meeting.

Table 4: Administrative kick-off meeting overview

Timing:	Soon after project registration and staffing have been completed.
Participants:	**SH** (organizer), LA, **Lead RSE** (optional). Others may be invited as needed.
Objective:	Introduce the collaboration with the Netherlands eScience Center, discuss the project setup, and identify any project-specific administrative or organizational requirements before project execution begins.
Duration:	1 hour.
Location:	Online.

Responsible: SH.

2.5 Kick-off meeting

Following the administrative kick-off meeting, the **Lead RSE** organizes the project kick-off meeting with the project team, comprising the eScience project team, the LA, and the LA's team. The purpose of the meeting is to establish a shared understanding of the project objectives, work plan, responsibilities, and expected deliverables before project execution begins.

Prior to the meeting, and based on their review of the project proposal, the **Lead RSE** and the eScience Center project team identify any scientific or technical aspects that should be discussed with the LA. These may include:

- the scientific objectives and expected project outcomes;
- the proposed work packages, milestones, timeline, and overall feasibility;

- technical challenges, assumptions, implementation risks, and whether additional technical expertise or consultation from other RSEs is required;
- software sustainability considerations and the initial Software Management Plan (SMP);
- dependencies on data, infrastructure, or external collaborators; and
- any issues requiring clarification before project execution starts.

The meeting is supported by a standardized project kick-off agenda available in the project portfolio under the *Templates projects* folder [8]. The **Lead RSE** uses this agenda to ensure that all standard topics are covered consistently across projects, while adapting the discussion to the specific needs of the project.

The agenda typically covers the following topics:

- Round of introductions;
- Project introduction and scientific objectives;
- Review of the project work plan;
- Agreements on the initial project planning and deliverables;
- Software sustainability and the Software Management Plan (SMP);
- Any other business.

The meeting outcomes, agreed action items, and any presentation material are stored in the corresponding project portfolio folder. The **Lead RSE** updates the project planning accordingly and prepares the project for the execution phase. These documents establish the initial project baseline and provide the starting point for the project log maintained during the execution phase. The Secretary may assist the **Lead RSE** with scheduling the meeting.

Table 5: Project kick-off meeting overview

Timing:	Shortly after the administrative kick-off meeting.
Participants:	**Lead RSE** (organizer), **SH**, LA, project team. Others may be invited as needed.
Objective:	Establish a shared understanding of the scientific objectives, work plan, responsibilities, and expected deliverables before project execution begins.
Duration:	1.5 hours.
Location:	At the eScience Center or the project location.

Note: The **Lead RSE**, responsible **SH**, and LA are expected to attend the meeting in person at the eScience Center or the project location. Other project team members may participate online if appropriate.

Responsible: Lead RSE.

2.6 Legal agreements

The eScience Center promotes open, reproducible science and open-source software development. Projects funded through the eScience Center are carried out under the eScience Center Terms and Conditions [10]. However, project partners may require additional legal agreements to grant access to their facilities, data, infrastructure, or other resources. These agreements may also contain provisions related to intellectual property, confidentiality, data protection, or information security.

RSEs must not sign any formal agreements without consulting the responsible **SH**. Such documents include, but are not limited to:

- Guest agreement
- Data sharing agreement
- Non-disclosure agreement
- Consortium agreement
- Collaboration agreement
- Intellectual property (IP) agreement

The **SH** reviews the proposed agreement, consulting the **Lead RSE** where project-specific input is required, and determines whether additional review is required. If needed, the **SH** consults the Policy & Compliance Officer (P&CO) before the agreement is signed. The final signed agreement is archived in the corresponding project portfolio.

Responsible: SH.

2.7 Technology plan

Every eScience project involves technological choices that influence software development, data management, reproducibility, and long-term sustainability. Following the project kick-off meeting, the **Lead RSE** prepares a technology plan in collaboration with the Lead Applicant and the project team. The plan documents the initial technological decisions for the project and provides a shared reference for project execution.

The technology plan typically describes:

- selected technologies and rationale;
- software architecture or implementation approach;
- expected software features and data outputs;
- quality standards;
- reproducibility and sustainability considerations;
- key technical assumptions or constraints.

The technology plan is maintained as a living document and may be updated as the project evolves to reflect significant technological decisions agreed between the **Lead RSE** and the Lead Applicant and project team. An example technology plan is available in [11].

The **Lead RSE** is encouraged to consult colleagues with relevant technical expertise when preparing the technology plan. Where appropriate, the Community Manager (CM) may advise on software dissemination, community engagement, and strategies to promote software reuse and adoption. Likewise, the eScience Center's Special Interest Groups (SIGs) may provide guidance on domain-specific technologies, software engineering practices, and other technical topics relevant to the project.

Table 6: Technology plan overview

Prepared by:	**Lead RSE**, in collaboration with the LA and the project team. Where appropriate, colleagues with relevant technical expertise, the Community Manager, and Special Interest Groups (SIGs) may provide advice.
Target audience:	Project team.
Timing:	Shortly after the project kick-off meeting and before substantial software development begins.
Objective:	Document the technological decisions agreed for the project, including the selected technologies, implementation approach, software quality considerations, and other technical decisions that guide project execution.
Format:	Free-format document, maintained as a living document throughout the project as needed.

Responsible: Lead RSE.

3 Project execution

Projects at the eScience Center range from 3 months to 5 years in a duration, depending on the call through which they were granted. In all call projects, the PM (supported by **Lead RSE**) monitors progress and involves relevant stakeholders as needed. The **Lead RSE** takes a central role during the execution phase, ensuring regular team meetings – typically monthly for the full team (varies with team size) and bi-weekly with the LA LA and/or LA team.

From a technical project management perspective, an eScience project is a research process that requires a flexible execution plan rather than a one-size-fits-all approach. The team must adapt to unexpected challenges and deviations from initial plans, if necessary. Therefore, during execution phase, the **Lead RSE** and RSEs are encouraged to utilize diverse project management and communication methods, such as those mentioned in Section 1.1, including Turing Way [12], minimalist project management [13], the SCRUM guide [14], ShapeUp [15], and other best practices [16, 17].

3.1 Project logging

eScience project team members routinely log important project events and agreements. The project log is placed in the Project portfolio. The **Lead RSE** keeps the log up to date (see example in Appendix D). The project log facilitates the information flow between different stakeholders about project activities.

The following should be included in the project log:

- RSD project page URL (see instructions for the RSD page in Appendix E)
- key meetings with dates, links to slides, and decisions
- infrastructure used and decisions regarding infrastructure
- workshop/conference participation (or this is registered as an output in RSD)
- management plan updates
- technology plan decisions and updates
- (Code) review outcomes.

Links pointing to other documents (e.g., files in the Project portfolio, project output, repositories) should be used in the project log to improve readability of the log and avoid duplicate information.

3.2 Status update meetings

The PM stays informed about the status of the project and communicates with the **Lead RSE** on a regular basis.

Scheduled:	Once every 4-6 weeks
Stakeholders:	PM (organizer), **Lead RSE**, optionally: TL [a], other RSEs.
Purpose:	Status update on the project and discussion around project management.
Duration:	30 min –- 1 hour
Location:	In-person meeting is default.

[a] The TL participation is mandatory for the technology-oriented projects.

PM and **Lead RSE** discuss:

- project status (including any changes in a project workplan)
- technological issues, with due consultation of TL, respective SIG, or other RSEs, if necessary

- changes in technology plan, technological choices (Section 2.7), management plans (Section 3.6). For any of these changes, TL presence is required
- synergies with other projects in the Center
- issues related to the budget, communication, staffing, etc.
- knowledge development and transfer, potential for software reuse, software sustainability.

The frequency and duration of these meetings are at the discretion of the PM and depend on factors such as the experience of the **Lead RSE** and the size of the team and/or the project.

In projects that have a stronger focus on technology (such as the eTEC, CIT, SS projects), the TL is involved in these meetings more frequently. For some projects, update meetings can be combined (e.g., for projects within the same Call) or organized in the context of a larger meeting (such as a SIG on a relevant topic). Together with the **Lead RSE**, the PM decides on the format of the status update meeting.

3.3 Project team meetings

To keep the entire project team informed on project progress, the **Lead RSE** together with the LA organizes a periodical project meeting. The frequency and format depend on the complexity of the project and size of the project team.

Scheduled:	Once every 2-6 weeks
Stakeholders:	**Lead RSE** or LA (organizer), LA (chair), RSEs and other project team members from the LA side.
Purpose:	Progress update on the project by all team members.
Duration:	30 min -- 1 hour
Location:	In-person meeting is default.

The agenda of this meeting should include:

- status update from all stakeholders
- discussion of scientific progress
- discussion of technological progress, issues and choices
- alignment of project progress with the project workplan, and adjustment of the latter, if necessary.

3.4 Writing hours and managing project budget

The PM and **Lead RSE** must have a firm grasp of the project budget and the project duration. This information is in the Awarding letter, the proposal and Exact.

The project execution phase roughly consists of three parts: exploration, development, sustaining. In call projects, the budget is roughly split as follows: 25% for exploration (including learning), 50% for development, and 25% for sustaining (including research and dissemination activities). The **Lead RSE** must consult the PM if project execution deviates from this plan. The left plot of Figure 2 assumes a constant budget spending rate. If possible, RSEs are encouraged to complete the first two phases faster, keeping the end date unchanged. This leaves more time for the sustaining phase (cf. Figure 2), which benefits outputs like publications. It also leads to more effective budget use under the current eScience Center financial system.

The LA and RSEs are free to spend the awarded hours ahead of the project end date. However, the **Lead RSE** is responsible for results being delivered on time for the project and not exceeding the budget, and timely informing the

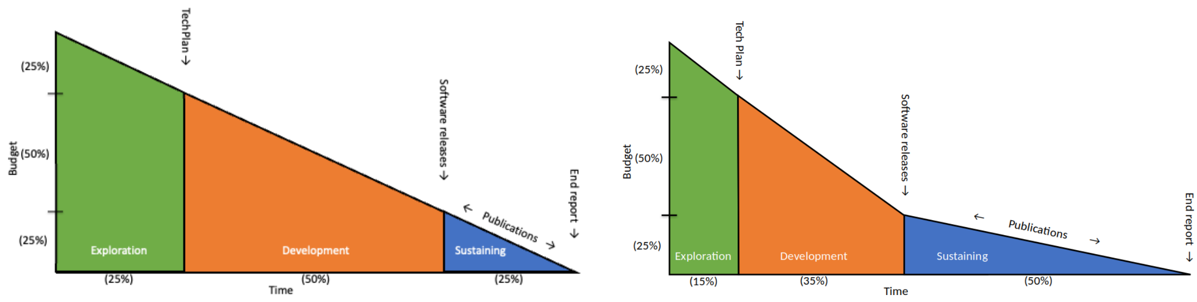

Figure 2: Project's budget breakdown: standard (left) and fast (right)

PM and the LA about any changes to the agreed planning. If a project budget is fully spent ahead of its schedule, the PM asks Finance to restrict writing on the project budget only to the PM.

The eScience project team (including PM and RSEs) must submit project hours in Exact by the end of each month. All entries must be completed within one working week after month-end. For regular call projects, RSEs can write hours once the project is active in Exact, typically within a month of grant approval. The PM records management hours on the project budget. Project hours are managed by different parties with different responsibilities:

Stakeholder	Responsibilities	More info
PM	• reviews and approves project hours by the 5th workday of the following month. • shares monthly hour status with RSEs via Ganttic • flags issues (e.g., incorrect budgets) to Finance • notifies Finance of budget changes.	
Lead RSE	• monitors project hour expenditure • signals deviation from the workplan to the PM	
Finance	• maintains accurate budget information • monitors and processes approved project hours • shares monthly financial info with budget holders (PMs). • recalculates project budget if extended past year threshold. • sets project to read-only if hours are depleted/exceeded early.	All budget changes require a formal decision (see 3.13).
DoT	• monitors and approves software sustainability hours	See the SS protocol [2]

It is possible to travel for a project, however RSEs must ask approval of their line managers (a respective **SH**) before committing to an event requiring travel and fill out a travel form. See the Intranet for more information.

For **external projects**, only the PM and RSEs formally working on the project may write hours. Some projects (e.g., Horizon Europe) allow only direct project activities – non-project activities like SIG involvement cannot be charged on the project. The **Lead RSE** and PM must know these restrictions or consult Finance if unsure.

3.5 Workshops

Some call projects require the LA to organize workshops. These workshops aim at fostering research communities around the software developed in projects. Workshops may focus on early user feedback, new features, or connecting existing tools to broader communities. Based on the call text, the PM and **Lead RSE** determine whether one of more workshops must be organized. Finance provides the project's workshop budget status. There are two types of workshops, namely, (a) organised by the LA, and (b) organized by the Lorentz Center.

For workshops organized by the LA: the LA writes and submits the workshop plan(s) to Finance in a timely manner (see [8] for templates). Finance approves the plans in consultation with the PD and PM. The **Lead RSE** is expected to contribute to the workshop and its organization. A CM can advise the LA (through the **Lead RSE**) on fostering engagement and growth of relevant software communities; a CM is involved in the introductory part of the workshop, including addressing attendees. Finance handles the administrative and financial aspects of the request and requests feedback on the scientific content from the PM team.

For workshops organised by the Lorentz Center: the LA must apply through the Lorentz Center webpage. The LA has the leading role in the application. The **Lead RSE** takes an advisory role in the writing and design of the workshop proposal and is expected to actively participate during the workshop event. The PM must ensure enough hours are allocated for the **Lead RSE** (or RSE from the project team) to help the LA with the proposal and attend the workshop, if they want to take an active role.

These workshops differ from the eScience-Lorentz Center Competition workshops that are jointly funded by the eScience and the Lorentz Center. Besides co-funding expenses, the eScience Center provides an additional in-kind support. The role of the eScience RSEs is described in the awarded proposals. The project initiation and the eScience team assignment follow the usual process. The team collaborates with the organizers on planning and outcome of the workshop (e.g. research paper, white paper, software release, or grant consortia). The **Lead RSE** plays a pivotal role in preparing and delivering the workshop, and, possibly, in publishing any outcome drafted during the event.

3.6 Data and Software Management Plans

For some projects, Data and Software Management Plans (DMP [18] and SMP [19]) outline how project data and software are maintained. Depending on the call, the LA must submit complete DMP and SMP documents within 6 months of the project. The **Lead RSE** may assist in drafting the DMP, which the LA submits to the PM, who asks a TL for review and approval. For call projects since 2021, SMPs are included in project proposals.

The LA is responsible for keeping both plans up to date and must inform the PM and TL of changes via the **Lead RSE**. The PM may request updates as needed, with support from the **Lead RSE**.

3.7 Knowledge transfer

To increase visibility of the project and its results, the project team (including RSEs, PM, TL), Communications, CMs, share knowledge and outcomes both inside and outside of the organization. The **Lead RSE** ensures that

- project results are timely shared with Communications, CMs, and relevant **SH**;
- project and software pages on the RSD are properly updated; and
- specific requests to facilitate project visibility are sent to Communications by RSEs.

Moreover, PMs, Lead RSEs and TLs work together to spot opportunities for cross project collaboration (e.g., by reusing software or knowledge in these projects or as a new reusability project, read more in Section 3.14.1).

3.7.1 Output management

Project deliverables include research articles, presentations, talks, posters, tutorials, datasets, blog posts, white papers, and workshops, as well as software-related outputs such as code releases, software papers, demos, videos, tutorials, and training materials. RSEs strive to apply FAIR principles [20, 21] to all project deliverables. They should have

- (concept) DOIs from publishers or open-access archives like Zenodo, arXiv, or DANS;

- acknowledge the eScience Center project grant [4]; and
- listed RSEs working on the project as (co-)authors;
- adhere to the GenAI policy [22].

RSEs must record all project outputs in relevant systems and databases (see Table 7) to ensure knowledge transfer. The PM expects the team to follow the deliverables plan outlined in the proposal and workplan. RSEs contribute to publications and software/data releases.

Table 7: Output management

What	Responsibility of	Responsible for	URL	Additional info
Zenodo (NLeSC community)	PM / CMs	curating and approving new publications	Zenodo link	• Publications on Zenodo are added to the community
	RSEs	getting a (concept) DOI for a data or software release, or a document (e.g., non-peer-reviewed publications)		
RSD, software and project pages	PM / **Lead RSE**	• ensure the eScience team has maintainer access • decide on reuse of the project page if the project continues under new funding.	See [23]	See Appendix E
	RSEs	• create project page, if needed • maintain pages with complete, up-to-date metadata		
Project portfolio	PM	adding a direct link to the project portfolio in Ganttic	cf. [8] for structure explanation.	An internal archive with regular automatic backups. Direct link to the portfolio folder available in Ganttic
	Lead RSE	ensuring that all docs are available		
	RSE	• upload outputs (e.g., papers, reports, presentations) • link the source material in the project log		
The eScience Center blog	RSEs	• Create a blog post about the project, its progress, or outcomes. • add final blog URL to project RSD page	Indexed at eScience Blog	Instructions on blogging are posted on the Intranet [2]
	Editorial Team	• Review blog post entry of project team • Help at any stage with writing and publishing process		

Open access publications and open source software are mandatory for all call projects. The PM and **Lead RSE** inform the LA if needed. Open access fees are budgeted by Finance annually in the call budget (with the PD approval); if no budget is available, alternatives can be explored:

- publishing through the LA's institution (either open access funds are available or the LA organization is connected to publish open access in a lot of journals through the library without a fee)

[4] By adding the following sentence into the publication: "<full project title> is a project of the Netherlands eScience Center, funded under grant number <grantid>."

- choosing the best option for open-access publishing (is it a diamond or gold open-access journal)
- publishing closed access and self-archiving either a preprint or publication after six months under Taverne agreement in Dutch copyright law (cf. [24, 25]).

The **Lead RSE** and PM consult Finance regarding payments for an open access publication.

For **external projects** the expected deliverables are also part of the formal project documents (proposal, contract, etc. The **Lead RSE** is expected to keep the PM informed of the status of deliverables throughout the project.

3.7.2 Outreach

The **Lead RSE** promotes the project's visibility through project demonstrators, presentations, and other means. All RSEs are expected to communicate externally about the project and its deliverables, as outlined in Section 3.7.1. Communications supports the project team by highlighting project through various channels, including but not limited to news items, social media posts, videos and interviews about relevant the scientific impact of the research software developed or used. The **Lead RSE** (or occasionally the PM) provides Communications with relevant information.

Blog posts are an optional but highly recommended output of the projects. Any team member, from LA to RSE, can author them. Topics may include simplified research summaries, tutorials on learned skills or technologies, or updates on workshops, publications, and other project outputs.

RSEs share project progress and results (e.g., technology plan, milestones, code releases) with colleagues, including SIG presentations. Each RSE prepares and updates a 3-slide deck or pitch [8]. The **Lead RSE** ensures a demonstrator is available after the first major software release.

The LA and their team are encouraged to participate in relevant Digital Skills Workshops [26] from the eScience Center. Furthermore, if the LA and their team require a project-specific training workshop, the **Lead RSE** involves

- workshop coordination with CMs, who can advise on organization and training material development; Finance handles any related payments if applicable;
- the PM discusses RSE hours spent on training organization and consults Finance if needed.

For **external projects**, consortium agreements (e.g., Non-Disclosure Agreement or NDA) may define result communication. The **Lead RSE** and PM review these at project start to clarify communication boundaries.

3.8 Code quality and sustainability checks

Ensuring software quality and sustainability is integral to the code development process cycle at the eScience Center. All RSEs are expected to follow our software development guide and best practices [27] and the GenAI policy [22]. Code should be designed to be as generic and reusable as possible from the start.

3.8.1 Code development

At the initial stage of code development in the project, the **Lead RSE** together with RSEs:

- set up a GitHub organization for the project, following the eScience Center Guide and the Turing Way [12]
- add the URL of this organization to the RSD project page.

RSEs must always ask at least one project team member, relevant SIG member or other RSE at the eScience Center to review, comment and approve pull requests in the project codebase.

3.8.2 Code review

As part of the annual review (see Section 3.9), a code review is organized by the **Lead RSE**. Depending on the project it could take the form of a reusabilithon[5], a review of code on GitHub, or something else entirely (format to be approved by TL). The reviewers for this process are typically other RSEs at the eScience Center.

The goal is to review software of the project for:

- its usability (reproducing steps of installations, and running it on a machine/laptop)
- overall software quality and suitability
- whenever appropriate, correctness of code
- adherence to the technology plan (Section 2.7)
- adherence to eScience Center best practices
- opportunities for reuse of software in other projects
- correct inclusion in output systems (Section 3.7.1).

Reviewers make written suggestions for improvements, and flag major issues encountered. These issues serve as input for the TL for the formal annual review meeting (see Section 3.9). These notes are stored in the project log.

3.9 Annual project review meeting

Scheduled:	Yearly
Stakeholders:	PM (organizer, chair), **Lead RSE**, LA, TL, optional: other project team members, **SH**
Purpose:	• to ensure that the project is still on track • to discuss any persistent issues to ensure optimal collaboration between project team • to explore opportunities beyond the project
Outcomes/Actions:	List of agreed actions and PM/TL advice for next steps.
Duration:	Max 1.5 hours
Location:	At the eScience Center or the project location[a]

[a] Mandatory participants must attend the meeting in person, while optional participants may join via video conference.

For call projects longer than one year, the PM organizes annual reviews. The details are described in the terms and conditions document [10]. For one-year projects, review meetings are optional and decided by the PM in consultation with the TL and **Lead RSE**.

Each review includes discussion and actions on reusability and sustainability of project results (as described in the DMP and SMP), status of collaboration, and follow-up opportunities. The agenda of this meeting is:

- introduction and purpose of meeting
- scientific goals status (the LA and team)
- project output status (the project team)
- collaboration status (including hours admin and bottlenecks)

[5] Coined by the Software Sustainability SIG, the term refers to a 2-3 hour session where a group of RSEs evaluates a piece of software for usability, provides feedback to the developers, and formulates recommendations to improve its (re)usability.

- use of digital infrastructure and SURF support (if applicable)
- next steps and future opportunities.

For **external projects**, review meetings are typically part of the project process. The PM, with the **Lead RSE**, decides if a (lightweight) internal review is needed.

3.9.1 Review meeting preparation

	Stakeholder
Prepared by:	LA and **Lead RSE**, with optional input from other stakeholders.
Reviewed by:	PM accountable for project, TL accountable for technology.
Target audience:	PMs, TLs, **SH** and RSEs.

The **Lead RSE** coordinates preparations for the review meeting:

- request the PM to share the standard review meeting slide template [8] with the LA team
- requests project team members to contribute to the review meeting slides, wherever appropriate,
- compiles technology status report (see Section 3.9.2).

LA and **Lead RSE** jointly prepare the slides:

- 3-5 slides summarizing progress toward research goals. The LA is not expected to present published content or the original workplan or proposal. If applicable, the LA reports on workshops.
- 1-2 slides on the technology plan status, including reuse, adoption, and sustainability, supporting the technology status report.
- highlight any scientific or technical bottlenecks and assess if the project is on track or needs replanning;
- the PM reports on the RSE hour status (i.e. the number of hours already spent); and
- the project team provides a feedback on collaboration and suggest improvements if needed.

Additionally, the **Lead RSE**:

- together with the LA, ensures all output is properly registered (see Section 3.7.1), and adds missing URLs/DOIs to the slides;
- coordinates with the team to finalize the presentation at least 2 working days before the review meeting;
- uploads the slides to the project portfolio; and
- informs PM and TL that slides are ready and are in the project portfolio.

3.9.2 Technology status report

Prior to the annual review meeting, the PM asks **Lead RSE** to fill in the technology status report. This report summarizes the technical progress of the project and includes RSD project page with its deliverables, URLs to project plans, software quality and community involvement. The **Lead RSE** prepares it in collaboration with the project team; it serves as input for the TL during the review. The **Lead RSE**:

- collects output and relevant information from LA and team

netherlands eScience center

- meets with CM (for the community outreach activities) to draft the technology status report
- draft the rest of the report with the TL

Two weeks prior to the review meeting, the **Lead RSE** submits the completed report to the TL. The TL conducts a code review or software health check based on the report and shares the results with the eScience team. If needed, the **Lead RSE**, TL, and PM may meet to discuss the findings before the annual review meeting.

The report and the review(s) are archived by the TL in the project portfolio.

3.9.3 At the review meeting

The time breakdown of the meeting agenda is follows:

- presentation by the LA (max. 20 min)
- presentation by the **Lead RSE** (max. 20 min), including the RSE roles and deliverables
- discussion (max. 40 min)
- summary, action points and conclusions.

The PM chairs the meeting and, with the TL, acts as reviewer. The TL highlights tech/software issues. Everyone contributes to the discussion. PM and TL assess deliverables, and comment on their status:

- have the objectives outlined in the proposal been sufficiently addressed? (PM)
- does the project follow the workplan in terms of deliverables? (PM)
- has the output been registered according to the rules of output management (Section 3.7.1)? (PM, TL)
- does all project output have publications (including software and data papers)? (PM)
- are there any issues flagged during the code review that needs to be discussed with the project team? (TL)
- does the project team sufficiently engage and align with relevant communities (e.g., via the workshops)? (PM)
- does the project adhere to the technology plan, SMP and DMP? (TL)

The eScience project team comments on any further possibilities for reusability, adoptability and sustainability of the software, and the project team comments on possible collaborations beyond the project.

The PM updates the slides with action points, agreements and plans (with the project partners agreement). The PM logs the meeting in the project log.

3.9.4 After the review meeting

PM shares the updated slide deck with the project team members to check the agreements written down. PM ensures that the final version of the presentation(s) uploaded to the project portfolio is correct.

3.10 Reporting

For call projects the annual review meeting and end report (Section 4) serve as formal progress reports.

External projects may require periodic reporting to the consortium on progress and deliverables. The PM and **Lead RSE** consult the agreements, contract, and workplan; Finance handles financial input. The external coordinator (e.g., EU project coordinator, NWO programme officer) signals deadlines on the report delivery. The **Lead RSE** drafts the eScience Center's contribution, the PM reviews it. Finance fills in the financial part of the report (signed by the DoO if needed). The PM submits the final report to the external project coordinator (via EU portal done by Finance) and archives it in the project portfolio.

3.11 Conflict resolution and complaint procedure

The eScience Center adheres to the Code of Conduct outlined in the Personnel Policy (cf. [28]). Conflicts involving eScience and the LA team on the projects should be addressed using this four-step process:

1. The **Lead RSE** first seeks resolution with the LA; the PM may be consulted or join in discussion if needed.
2. If unresolved, the issue is escalated to the PM (via the **Lead RSE**), who organizes a meeting to mediate.
3. Persistent issues can be escalated to the PD via a written summary submitted through the PM.
4. The PD may escalate to the DT if necessary.

If the issue concerns the PM, RSEs may escalate to their manager (**SH**). An external confidential advisor ('Vertrouwenspersoon') is also available for anonymous support [2]. Conflict resolution may lead to project adjustments, such as staffing changes (cf. Section **??**).

3.12 Non-RSE Consultants

Certain issues require consultation beyond the project team. Team members must inform the PM, who may delegate actions as needed:

- GDPR/personal data: consult the GDPR contact [2].
- Software/data quality, access and sustainability: contact TL and CMs.
- Scientific integrity: contact the integrity officer [2].
- SURF-related matters: contact SURF liaisons [2].
- IP/licensing: contact TLs and DoT. (cf. Section 2.5).
- Legal matters: contact the DoO (cf. Section 2.6).
- External funding/opportunities: contact the PD (cf. Section 3.14.3).

3.13 Changes to the project

During the project life cycle, the workplan may change substantially:

- New deliverables because of additional funding
- New workplan because of changes in the research goal and/or in the technology used
- Timeline, leading to a different end date.

Any of these changes needs explicit approval from the PM team or the DT.

Type of request	Decided by:
Budget requests within the PM mandate [2]	PM team
All requests regarding budget changes outside the PM mandate	DT (via PD)
Early termination	DT (and DT informs the Board)

For **external projects**, changes must follow the formal agreements (e.g., grant or consortium agreement). The PM handles extensions within their mandate or escalates to the DT. The PM informs the external funder or consortium of the outcome.

3.13.1 Proposal changes request

The LA must submit a formal request to the PM team (by email via the PM, in PDF format, signed) containing:

- project title and project number
- requested change (e.g., time/dates, RSE hours, scientific goal) and motivation for this change
- conditions such as deliverables: any new deliverables should have updated timeline. If none, state explicitly.
- any motivated budget change, such as
  - LA wants to increase their involvement
  - change in research personnel (if applicable in the case of older projects)
  - transfer between hardware and PYR for RSE or LA personnel costs (if applicable, for older projects)
  - any in-kind to cash change, or vice versa (including requests with the extra cash budget from the LA).
- any prior or planned inactivity on the project, such as
  - personnel shortage on the LA side due to maternity/sick leave or hiring delays (but there is a concise timeline on the hiring procedure)
  - unavailability of RSEs
  - additional data that needs to be collected.
- any delay with the start date.

3.13.2 Processing the changes request and decision

Upon receipt of the request, the PM assesses if the request should be granted based on considerations such as

- whether the new objective is scientifically promising or technologically interesting? (if applicable)
- collaboration status with the LA;
- prior problems regarding the project;
- benefits for the eScience Center (e.g., smooth wrap-up, future funding opportunity); and/or
- availability of RSEs with relevant expertise to work on it.

The PM can consult with RSEs and the TL on whether the new planning is feasible, and with Finance for a budget remaining after necessary recalculation. In case of additional funding, the DT (via the PD) will decide, after a budget calculation by the Finance and approval by the DoO. Otherwise, the PM puts the request on the agenda for the next PM meeting, containing:

- the motivated request (uploaded to the project portfolio
- the recommended action
- the prepared decision on the PM meeting agenda.

The PM team may request more information from the LA via the PM (and thus postpone the decision on the request). The LA can provide the new information via an additional PDF signed letter or as amendment to the original letter.

After the final decision, the PM notes the official decision in the decision document. If the request is not approved, the PM communicates this to the LA. If the request is approved, the PM

- coordinates with Finance to finalize the extension (budget/Exact update, extension letter, LA communication);

- communicates the extension to the **Lead RSE**.

The **Lead RSE** then

- communicates the extension to the project team;
- updates planning and adjusts staffing, if necessary;
- ensures website and RSD are updated (e.g., if dates or affiliation changed); and

If the request involves a DT decision, the PM submits a request formally through the PD.

3.13.3 Early project termination

Early project termination may occur by mutual agreement, or be initiated by the LA or the eScience Center. In the first two cases, the LA submits a signed letter (via the PM) to the DT, requesting project termination, explaining the situation and proposing how to handle remaining project resources (e.g., RSE hours, cash contribution, FTE commitment of the LA, workshops, sustainability budget).

The PM may request project termination if the LA or project partners violate the terms of Awarding Letter or Special conditions ("Bijzondere Voorwaarden"). The PM submits a letter with the explanation to the DT via the PD. If approved, the PM informs Finance, which finalizes the process (by making changes in Exact and preparing a termination letter).

3.14 Opportunities beyond the project

A project team can explore different opportunities for ideas that stem from the project that go beyond its scope and/or budget. Appendix **??** summarizes the role of PM in such projects. The PM discusses these opportunities with the project team during the review meeting.

3.14.1 Increasing reusability (in this document called software sustainability)

Some projects or calls have dedicated software sustainability budgets [29]. Until 2020, each project received its own Generalization ("Generalisatie") budget for reuse of project results. Since 2020, this budget is allocated at the programme or call level. The PM and **Lead RSE** must check the call text, awarding letter to confirm its availability.

The PM signals potential for reuse to the TL, who discusses it with the **Lead RSE** and, if needed, the TL team. RSEs with ideas for sustainability projects can contact the TL or DoT. The process follows the SS protocol [2].

3.14.2 Knowledge and Development

For the development of broad and deep knowledge on digital technologies and their application, RSEs can apply for so-called Knowledge and Development (KD) project. The process is described by the KD protocol [2].

3.14.3 External funding

The project team may decide to pursue other funding opportunities. The eScience Center encourages RSEs to pursue external funding opportunities to promote the organization profile as research organization. PD oversees the Acquisition activities and the process. The relevant information is available via Intranet [2].

3.14.4 Fellowship

To stimulate community engagement lasting longer than project lifetime, the eScience Center funds annual Fellowship Program [30, 31]. The eScience project team is not eligible for this program, however, the LA and their team are.

4 Project closing

Project closing is the final phase of a project. In this phase the **Lead RSE** requests and previews the end report, the PM (with the help of Finance) processes the end report and accepts the project deliverables. Once the project is formally closed, RSEs can no longer write hours or work on this project.

4.1 Handling end report

All completed call projects at the eScience Center must have an end project report.

Written by:	the LA, assisted by the **Lead RSE**.
Target audience:	PMs, RSEs, Communications (layman summary), Finance (accountants), TLs
Schedule:	• written in last months of the project, • submitted 3 months after the project end at latest, • archived in the project portfolio.
Approved by:	The PM team and Finance.

For call projects, one month before the project end, the **Lead RSE** requests the LA to submit the scientific and financial end report ('Financieel en wetenschappelijk eindverslag'), providing the eScience Center template [8].

RSEs provide the LA with all necessary information for the end report.A complete end report includes:

- Public summary (in English) with objectives and results,
- Challenges (the team encountered during the project),
- Opportunities (the work of the project led to),
- Remarks on sustainability and latest management plans,
- Outcomes missing from the RSD (e.g., software, papers, presentations, workshops, testimonials),
- Information on the project workshops,
- Financial overview (if applicable),
- LA's signature and date (or financial/project manager's if LA unavailable).

For collaborative calls, the LA and **Lead RSE**'s report to other funders (e.g., NWO) suffices if it covers the above points.

Before the formal closing of the project, the **Lead RSE**:

- receives the signed end report from the LA (preferrably, in PDF format)
- verifies the scientific part of the end report against the end report checklist. If needed, the **Lead RSE** asks the LA for additional information or corrections to the end report, before submitting it to the PM.
- ensures all missing project output is registered in the appropriate systems (see Section 3.7.1)
- archives the end report, the lastest DMP and SMP, checklist in the respective project portfolio.

If the end report is satisfactory and all the aforementioned steps are completed, the **Lead RSE** submits it to the PM to get the project formally closed. The PM requests Finance to review the financial part of the report and either approve it or request corrections to it.

Projects that are funded by software sustainability budgets have their own procedure for end reports (see Appendix **??**). The end report of these projects consists of output registered on the RSD project page and updated summary of the project.

For **external projects** the way a project is formally closed depends on the formal documentation for a project and requirements from the external funding organization. If an end report is required, the **Lead RSE** contributes to it (PMs can assist when needed), and care should be taken to reserve some time (and budget) during the project for this effort. Finance prepares the financial part of the report. The **Lead RSE** shares the final version of the end report written for the external party or funding organization (e.g., EU, NWO) with the PM and Finance, and archives it in the project portfolio. In any case, the **Lead RSE** ensures that the RSD project page is up-to-date (with complete deliverables and layman summary of the completed project).

Finance periodically sends a list of missing end reports to the PM team. If the end report is not submitted yet, the PM sends a reminder to the LA.

4.2 Formally closing the project

If both scientific and financial part of the end report is satisfactory, the PM puts decision to formally close project on the PM meeting agenda. After the formal decision[6], the PM notifies the TLs, Finance and Communications (with the links to the documents). Finance handles the approved reports and formalities related to closing the project. This includes getting the final signature by DoO or Executive Director on the official letter for the LA about the project closing ('Afsluitingsbrief'), communicating it to the LA, and archiving the letter in the project portfolio.

The PM ensures that the project is marked as complete on the

- RSD: The **Lead RSE** updates the project page with the lay summary (from the end report), any missing meta-data (e-Infra use, keywords, deliverables, etc.) and sets the project status to 'Closed'. See also Appendix E.
- Corporate page: Communications may also request more information to promote the completion of the project through various channels, including but not limited to a news item, social media post, video and interview.
- Exact/Project portfolio: Finance closes the Exact project budget, uploads the official closing letter to the LA, as well as all appropriated documents.
- Ganttic: The PM ensures that no one is assigned to the project in the future.

Related links and references

[1] The Netherlands eScience Center. 2015 NLeSC Protocol voor Calls, Subsidietoekenning en Projectmanagement.

[2] The Netherlands eScience Center. Intranet. http://intranet.esciencecenter.nl/.

[3] The Netherlands eScience Center. What is a Research Software Engineer? A definition by the Netherlands eScience Center. https://doi.org/10.5281/zenodo.7994286, 2023.

[4] The Netherlands eScience Center. Research Software Engineer at the Netherlands eScience Center: Job Description. https://doi.org/10.5281/zenodo.7805870, 2023.

[5] GDPR.EU. General Data Protection Regulation (GDPR). https://gdpr.eu/. Accessed: July 29 2026.

[6] Digital Competence Center Praktijkgericht Onderzoek (DCC-PO). AVG en privacy in onderzoek. https://dcc-po.nl/kennis-expertise/themas/juridisch/avg-en-privacy-in-onderzoek/, 2025.

[6] Formally recorded by the PM team, and thereafter ratified by the DT team (see more in [2]).

[7] Universiteiten van Nederland. Handreikingen voor gebruik van persoonsgegevens in wetenschappelijk onderzoek. https://www.universiteitenvannederland.nl/handreikingen-voor-gebruik-van-persoonsgegevens-in-wetenschappelijk-onderzoek, 2025.

[8] The Netherlands eScience Center. Project Portfolio (the internal project repository).

[9] Carlos Martinez-Ortiz, Rena Bakhshi, Yifat Dzigan, Nicolas Renaud, Faruk Diblen, Berend Weel, Maarten van Meersbergen, Niels Drost, Sven van der Burg, and Fakhereh Alidoost. Structured and Unstructured Teams for Research Software Development at The Netherlands eScience Center. *Computing in Science & Engineering*, 24(3):25–32, 2022. doi:10.1109/MCSE.2022.3167448.

[10] The Netherlands eScience Center. Bijzondere voorwaarden Netherlands eScience Center subsidies. Available via https://www.esciencecenter.nl/calls-for-proposals/. Accessed: July 29, 2026.

[11] J. H. Spaaks. ASReview & HOLM plan. https://doi.org/10.5281/zenodo.7849050, April 2023. Zenodo. doi:10.5281/zenodo.7849050.

[12] The Turing Way Community. The Turing Way: A handbook for reproducible, ethical and collaborative research, April 2025. doi:10.5281/zenodo.3233853.

[13] OMIMO. micro.p3.express, minimalist project management for micro projects. https://micro.p3.express/, 2023.

[14] Jeff Sutherland and Ken Schwaber. The Scrum Guide. https://scrumguides.org/index.html.

[15] AgileFirst. Shape up: A complete guide to this new development methodology. https://agilefirst.io/what-is-shape-up/, 2024.

[16] Kent Beck and Cynthia Andres. *Extreme Programming Explained: Embrace Change*. Addison-Wesley Professional, 2nd edition, 2004. ISBN: 978-0321278654.

[17] Greg Wilson, DA Aruliah, CT Brown, NPC Hong, M Davis, RT Guy, SHD Haddock, KD Huff, IM Mitchell, MD Plumbley, et al. Best practices for scientific computing. *PLoS biology*, 12(1):e1001745, 2014. doi:10.1371/journal.pbio.1001745.

[18] Maria Cruz and Eveline van den Berg. NWO DMP assessment rubric, January 2020. doi:10.5281/zenodo.3629157.

[19] Carlos Martinez-Ortiz, Paula Martinez Lavanchy, Laurents Sesink, Brett G. Olivier, James Meakin, Maaike de Jong, and Maria Cruz. Practical guide to Software Management Plans, October 2022. doi:10.5281/zenodo.7248877.

[20] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, and *others*. The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data*, 3(1):160018, Mar 2016. doi:10.1038/sdata.2016.18.

[21] Neil P. Chue Hong, Daniel S. Katz, Michelle Barker, Anna-Lena Lamprecht, Carlos Martinez, Fotis E. Psomopoulos, Jen Harrow, and RDA FAIR4RS WG. FAIR Principles for Research Software (FAIR4RS Principles), June 2022. doi:10.15497/RDA00068.

[22] Netherlands eScience Center. Policy concerning the use of Generative AI at the Netherlands eScience Center (0.92). https://doi.org/10.5281/zenodo.10118669, 2023. Zenodo. doi:10.5281/zenodo.10118669.

[23] Research Software Directory. User documentation, 2025. URL: https://research-software-directory.org/documentation/users/.

[24] Universiteitsbibliotheek Utrecht. Amendement taverne. https://www.uu.nl/universiteitsbibliotheek/taverne.

[25] Open Universiteit. Taverne – Article 25FA – End user agreement. https://www.ou.nl/taverne-agreement.

[26] The Netherlands eScience Center. eScience Training: Digital skills workshops. https://www.esciencecenter.nl/digital-skills/.

[27] Niels Drost, Jurriaan H. Spaaks, Bouwe Andela, Lourens Veen, Janneke M. van der Zwaan, Stefan Verhoeven, Patrick Bos, Mateusz Kuzak, Ben van Werkhoven, Jisk Attema, Johannes Hidding, Vincent Hees, Carlos Martinez-Ortiz, Hanno Spreeuw, Joris Borgdorff, Katrin Leinweber, Faruk Diblen, Gijs van den Oord, Romulo Goncalves, Arnold Kuzniar, Dafne van Kuppevelt, Berend Weel, Christiaan Meijer, Jason Maassen, Pablo Rodríguez-Sánchez, Tom Klaver, Willem Robert van Hage, Felipe Zapata, and Tom Bakker. *Netherlands eScience Center - Software Development Guide*. Zenodo, September 2020. doi:10.5281/zenodo.4020564.

[28] The Netherlands eScience Center. CAO & Personnel policy. CAO is available via https://www.wvoi.nl/.

[29] Netherlands eScience Center. Software Sustainability at the Netherlands eScience Center (Version v5). https://doi.org/10.5281/zenodo.15646828, 2024. Zenodo. doi:10.5281/zenodo.15646828.

[30] The Netherlands eScience Center. Fellowship Programme. https://www.esciencecenter.nl/fellowship-programme/.

[31] Netherlands eScience Center and de Boer, L. and Martinez-Ortiz, C. eScience Center Fellowship. https://doi.org/10.5281/zenodo.15653175, 2024. Zenodo. doi:10.5281/zenodo.15653175.

Appendices

A Applying the protocol to external projects

Projects funded by external organisations (e.g., NWO, Horizon Europe, charities, industry, or public-private partnerships) are subject to contractual obligations defined by the funding body or collaboration agreement. Where these obligations differ from the guidance provided in this protocol, the external contractual requirements take precedence. This appendix describes how the Project Management Protocol should be applied to externally funded projects and highlights the main adaptations required to comply with external funding conditions while maintaining the eScience Center's internal project management practices.

1. External agreements take precedence * Grant Agreement * Consortium Agreement * Customer contract * Project-specific governance 2. What remains unchanged * Roles at the eScience Center * Internal reviews * Documentation * Quality assurance * Risk management * Project closure 3. Typical differences * Additional reporting * Deliverables * Milestones * Steering committees * Consortium meetings * External approvals 4. Examples * NWO * Horizon Europe * Industry projects

B Applying the protocol to internal projects

C Lead RSE role description

This role description includes guidelines that need to be followed by RSEs fulfilling the role; they will be complemented by protocols.

1. Role	**Lead RSE**
2. Place in the organization	Project
3. Contacts	**SH**, project team RSEs, external project partners
4. Purpose	To be responsible for the day-to-day running of a research project at the eScience Center and to act as the main point of contact for the project. Each project has one **Lead RSE**.
5. Main tasks & responsibilities	• Agreeing with the accountable **SH** on the division of tasks and the allocation of time within the project • Coordinating day-to-day activities with other RSEs working on the project • Making sure that agreed activities, procedures and targets are carried out and met on time • Monitoring project progress, including project hour expenditure, and regularly reporting progress to the accountable **SH** • Ensuring that major technological decisions are appropriately reviewed and monitoring their implementation, in consultation with the accountable **SH** • Ensuring that generalization and re-usability opportunities are considered and implemented from the start of the project, in consultation with the accountable **SH** • Ensuring the presence of the accountable **SH** project at meetings where their participation is required (e.g. project kick-off and review meetings) • Solving everyday technical and managerial problems and escalating them to the accountable **SH** when necessary • Making sure all project outputs are properly released, documented and archived in the designated systems • Ensuring the visibility of the project through project demonstrators, slide decks and other appropriate dissemination activities
6. Competencies	• Leading • Project management • Communicating • Cooperating • Result orientation • Decision making
7. Available resources (budget, hours, training)	In project budget
8. How to get this role	**SH** assigns **Lead RSE** based on skills, experience, knowledge, interest and availability

netherlands eScience center

D Example of the project log

Running log for Project XXX

2026-03-12 Output: submitted paper

2026-02-02 Mr. X assigned as **Lead RSE**

2025-09-01 Kick-off meeting

Present – NLesC: AB (PM), AA (**Lead RSE**), AC (TL), AD (RSE)

Present – Team: FA (LA, TU Delft), PA (PhD student, TU Delft), RA (TU Delft)

Agreements:

- Lorem ipsum dolor sit amet, consectetur adipiscing elit,
- sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.
- Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris.

2025-11-10 SURF proposal granted

The proposal was approved by SURF. Access was provided to RC instead of Snellius, as it was considered sufficient for the project's computational and storage requirements.

2025-11-02 SURF infrastructure proposal

A proposal for SURF computing infrastructure was submitted after evaluating the project's computational and storage requirements.

Project Start

	Date	Slides/Meeting notes/URLs?	Notes
Administrative Start meeting	2025-09-01		
Project Kick-off	2025-09-01		
Review1	2026-06-01		To plan final date
Review2			
Tech plan v1	2025-09-20	github.com/shico/techplan.rst	TL comments...
Paper			TODO: Add to RSD
Presentation		URL (to the project portfolio)	Uploaded to RSD and Zenodo

E RSD project pages

Each project at the eScience Center has an RSD page.

Project Initiation At the project start, the following data should be added to the RSD project page:

- grant ID: currently, the Exact number of the call project or the Awarding letter reference (will be replaced by RAiD later). External funders (e.g., EU, NWO) often provide their own ID.
- funder(s): list of all funding organizations, including the eScience Center.
- dates: Project start and end dates.
- research domain: relevant project domain, often from the proposal. Multiple domains can be chosen.
- participating organizations: list of the organizations, including the eScience section(s).
- description: Typically copied from the proposal abstract.
- a cover image: initially a placeholder image provided by the Communication
- relevant keywords and categories
- the call type: the specific funding scheme or program through which the project is supported.

When reusing a project page under new funding, **Lead RSE** updates it with the new funding source and details.

Project Execution During the project, the following data should be added or updated as needed:

- repository link: URL to the GitHub (or other version control platform) containing the project's codebase.
- project team: the list of all project team members including the LAs and their team, with roles and ORCIDs where possible. RSEs should update the list as the team members change, and use the role field to indicate involvement periods.
- related projects: the link(s) to any precursor project(s) in RSD.
- deliverables: any project deliverables (see also Section 3.7.1) including but not limited to
  - all research deliverables of the project, i.e. publications such as papers or published datasets (using the DOI provided by the publisher), preprints of papers (using the DOI provided by the preprint server) self published research outputs, such as datasets, presentations, posters, reports, etc. (using the DOI it receives when uploaded to Zenodo), Online material such as websites, notebooks, online tutorials, blogs, videos, news items, interviews, etc. (registered using the URL and a short description).
  - all software deliverables: all reusable software should have its own RSD software page and be linked as related software. Single-use software (e.g., scripts) can also be archived or released via Zenodo and can be added via DOI. Versioned releases archived in Zenodo can be added via DOI, in cases when software has been developed in multiple projects.
  - design documents: the documents such as the Technology Plan, SMP and DMP can be published on Zenodo, and added using their DOIs.

Project Closing Once the project is coming to an end, in addition to the end report data, the following data RSEs are highly encouraged to add:

- testimonials from the project partners or user community
- list of all digital infrastructure used by the project team during the entire project

netherlands eScience center

F Internal Procedure of the Project Workshops Approval

The PM sends the LA a workshop proposal template prior to submission. The internal (assessment) procedure consists of several steps, as follows:

1. The LA submits the workshop proposal by email to the project PM. The submitted proposal consists of both scientific part and a financial part.
2. The PM provides the workshop proposal to Finance and DoO; they only assess the financial part. The DoO approves or rejects the financial part of the proposal and immediately informs the PM.
3. The PM only assesses the scientific part of the proposal and gives positive or negative advice to the PD. The PD approves or rejects the scientific part of the proposal and immediately informs the PM.
4. If a part of the proposal is not approved, the PM will contact the LA with a request for adjustment and/or additional information.
5. If both parts of the proposal are approved,
    - the PM archives the approved workshop proposal in the project portfolio folder.
    - The PM informs Finance of the overall positive decision with the request to process it in the financial system.
    - The PM prepares a letter to the LA informing that the workshop proposal has been approved. The letter also contains all further preconditions imposed on the LA and the organization of the workshop.
    - The PM asks the PD to sign the letter, saves the signed copy in the project portfolio folder, informs Finance, and sends a copy to the LA.

Key eScience Roles appearance

Lead RSE, 4, 9–23, 25–27, 32–34

SH, 4, 6, 8–12, 16, 17, 20, 21, 23, 32